

INDUSTRIAL ORIENTED MINI PROJECT REPORT ON

SENTIMENT ANALYSIS OF TWITTER DATA

Submitted in partial fulfillment of the requirements for the award of the degree of

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By

P. Sai Nithin	(23AG1A67B2)
M. Akshitha	(23AG1A67A7)
G. Venkataramana	(23AG1A6787)
G. Abhiram	(23AG1A6790)
B. Sai Charan	(23AG1A6771)

Under the guidance of

Mr. Peesala Ilanna



ACE Engineering College

Ankushapur(V), Ghatkesar(M), Medchal Dist - 501 301

(An Autonomous Institution, Affiliated to JNTUH, Hyderabad)

www.aceec.ac.in

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DEPARTMENT OF CSE (DATASCIENCE)

CERTIFICATE

This is to certify that the Industrial Oriented Mini Project Report entitled “SENTIMENT ANALYSIS OF TWITTER DATA” is a bonafide work done by P. Sai Nithin (23AG1A67B2), M.Akshitha (23AG1A67A7), G.Venkataramana (23AG1A6787), G.Abhiram(23AG1A6790), B.Sai Charan (23AG1A6771) in partial fulfillment for the award of Degree of BACHELOR OF TECHNOLOGY in CSE (Data Science) from JNTUH University,Hyderabad during the academic year 2025-2026.This record of bonafide work carried out by them under our guidance and supervision.

The results embodied in this report have not been submitted by the student to any other University or Institution for the award of any degree or diploma.

Guide name
Mr. Peesala Ilanna
Assistant Professor
Supervisor

Dr. R. Suresh
Professor
H.O.D., CSE-DS

External

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Pamulaparthi Sai Nithin	(23AG1A67B2)
Muniganam Akshitha	(23AG1A67A7)
Godishala Venkataramana	(23AG1A6787)
Gundoju Abhiram	(23AG1A6790)
Bodiga Sai Charan	(23AG1A6771)

SENTIMENT ANALYSIS OF TWITTER DATA

ABSTRACT

Sentiment Analysis of Twitter Data is a technique used to automatically identify and classify opinions expressed in tweets into different sentiment categories such as positive, negative, and neutral. With the rapid growth of social media platforms, Twitter has become an important source for understanding public opinion, trends, and user feedback. This project focuses on analyzing Twitter data using Natural Language Processing (NLP) and machine learning techniques to determine the sentiment expressed in tweets.

The system begins by collecting Twitter data and performing preprocessing steps such as removing special characters, stop words, URLs, and unnecessary symbols to improve the quality of the text data. After preprocessing, feature extraction techniques such as Term Frequency–Inverse Document Frequency (TF-IDF) are applied to convert textual data into numerical representations that can be processed by machine learning algorithms. A classification model, such as Random Forest, is then trained on the processed dataset to identify and categorize sentiments in the tweets.

The developed model is evaluated using performance metrics such as accuracy, precision, recall, and confusion matrix to measure the effectiveness of sentiment classification. The results demonstrate how machine learning techniques can effectively analyze large volumes of Twitter data and extract meaningful insights about public opinions. This project can be useful in various domains such as marketing analysis, product feedback evaluation, political opinion monitoring, and social media trend analysis.

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LIST OF ABBREVIATIONS

ABBREVIATIONS

NLP
ML
DL
API
SVM
CNN
RNN
LSTM
NB
LR
OS
RAM
CPU
IDE
URL

FULLFORM

Natural Language Processing
Machine Learning
Deep Learning
Application Programming Interface
Support Vector Machine
Convolutional Neural Network
Recurrent Neural Network
Long Short-Term Memory
Naïve Bayes
Logistic Regression
Operating System
Random Access Memory
Central Processing Unit
Integrated Development Environment
Uniform Resource Locator

I.INTRODUCTION

Social media platforms have become an important medium for people to share their opinions, thoughts, and experiences. Among these platforms, Twitter is one of the most widely used microblogging platforms where millions of users post tweets every day about various topics such as products, services, politics, events, and social issues. These tweets contain valuable information that reflects public opinion and sentiment.

Sentiment Analysis, also known as opinion mining, is a technique used to analyze text data and determine the sentiment expressed by users. It classifies the text into different categories such as positive, negative, or neutral. By analyzing Twitter data, organizations and researchers can understand public reactions, customer feedback, and trending topics more effectively.

This project focuses on analyzing Twitter data using Natural Language Processing (NLP) and machine learning techniques. The system collects tweets, preprocesses the data by removing unwanted elements such as stop words, special characters, and URLs, and converts the text into numerical features using techniques like TF-IDF. A machine learning model such as Random Forest is then used to classify the sentiments of the tweets. The results help in identifying patterns in public opinion and can assist organizations in making better decisions.

1.1 MOTIVATION

With the rapid growth of social media platforms like Twitter, people constantly share their opinions, reviews, and experiences online. This massive amount of data provides valuable insights into public sentiment regarding products, services, events, and social issues.

However, manually analyzing thousands of tweets is time-consuming and inefficient. There is a strong need for an automated system that can quickly process and analyze this data.

The motivation behind this project is:

- To **automate the analysis of public opinions** from Twitter data.
- To help businesses and organizations **understand customer feedback**.

- To identify trends and patterns in user sentiments.
- To provide **real-time insights** for better decision-making.

This project aims to leverage machine learning techniques to efficiently classify sentiments and extract meaningful information from unstructured text data.

1.2 PROBLEM STATEMENT

Twitter contains a vast amount of unstructured textual data that reflects people's opinions and emotions. However, extracting meaningful insights from this data is challenging due to its large volume, informal language, abbreviations, and presence of noise such as emojis, hashtags, and links. The main problem addressed in this project is how to efficiently process and analyze Twitter data to accurately classify the sentiment of tweets into categories such as positive, negative, or neutral using machine learning techniques. In today's digital world, a huge volume of unstructured textual data is generated every second on platforms like Twitter. Extracting useful insights from this data is a challenging task due to:

- The **large volume and high velocity** of tweets.
- Presence of **noise**, such as slang, abbreviations, emojis, and hashtags.
- Difficulty in understanding **context and sentiment polarity**.
- Lack of efficient tools to perform **real-time sentiment classification**.

The problem addressed in this project is:

“To design and develop a system that can automatically collect, preprocess, and analyze Twitter data to accurately classify sentiments into positive, negative, and neutral categories.”

The system should:

- Handle noisy and unstructured data effectively.
- Provide accurate sentiment classification.

1.3 OBJECTIVE

The main objectives of this project are:

1. To collect Twitter data
 - Gather tweets using keywords, hashtags, or datasets from Twitter.
2. To preprocess textual data
 - Clean tweets by removing noise such as URLs, mentions, hashtags, and special characters.
3. To extract meaningful features
 - Convert text into numerical form using techniques like TF-IDF.
4. To build a sentiment classification model
 - Classify tweets into positive, negative, and neutral categories using machine learning algorithms.
5. To analyze public opinion
 - Understand user sentiments about products, services, or events.
6. To visualize results
 - Present sentiment distribution using graphs like bar charts or pie charts.
7. To improve decision-making
 - Help businesses and organizations make better decisions based on user feedback.

1.4 LIMITATIONS

Although sentiment analysis provides useful insights, the system has certain limitations. Twitter data often contains slang, abbreviations, emojis, and sarcasm, which can be difficult for machine learning models to interpret accurately. The accuracy of the model also depends on the quality and size of the dataset used for training. Additionally, the system mainly focuses on text-based sentiment and may not fully capture complex emotions or contextual meanings present in tweets.

1. The system may fail to correctly detect sarcasm or irony in tweets.
2. Accuracy may reduce due to slang, abbreviations, and informal language.
3. Short tweets may lead to lack of context, affecting classification.
4. The model performance depends on the quality and size of training data.

II. LITERATURE SURVEY

Sentiment analysis of Twitter data has become an important research area due to the increasing use of social media for expressing opinions. Various approaches have been proposed over time to improve the accuracy and efficiency of sentiment classification.

Initially, researchers used lexicon-based approaches, which rely on predefined dictionaries of positive and negative words. These methods are simple and do not require training data, but they are limited in handling context, sarcasm, and informal language commonly found in tweets.

To overcome these limitations, machine learning techniques such as Naïve Bayes, Support Vector Machines (SVM), and Logistic Regression were introduced. These models learn from labeled datasets and provide better accuracy compared to lexicon-based methods. Feature extraction techniques like TF-IDF were commonly used to convert text into numerical form. However, these methods still struggle with understanding the context and sequential nature of language.

With advancements in technology, deep learning models such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Convolutional Neural Networks (CNN) have been widely used. These models are capable of capturing contextual and sequential information in text, leading to improved performance in sentiment classification tasks.

More recently, transformer-based models like BERT (Bidirectional Encoder Representations from Transformers) have achieved state-of-the-art results in sentiment analysis. These models understand the context of words more effectively by processing text in both directions, making them highly suitable for analyzing complex Twitter data.

Overall, the literature shows a transition from simple rule-based methods to advanced deep learning techniques. Modern approaches provide higher accuracy and better handling of informal language, making them more effective for sentiment analysis of Twitter data.

Title Name	Author Names	Technologies Used	Limitations	Year
Twitter (X) as an Electoral Barometer: Systematic Evidence from Sentiment Analysis of Twitter Data	M. Asokere, A. Wusu, O. Olabanjo	Machine Learning, NLP, Sentiment Classification	Limited validation with real election results; dataset bias	2025
Twitter Sentiment Analysis Using Machine Learning and Deep Learning Techniques	Safa Khalid Al Sarayrah et al.	SVM, Logistic Regression, LSTM, Deep Learning	Difficulty handling sarcasm and emojis; computational cost	2025
Sentiment Analysis: A Hybrid Approach on Twitter Data	C. G. Maurya	Hybrid ML, Opinion Mining, NLP	Complex processing pipeline	2024
BERT Model Adoption for Sarcasm Detection on Twitter Data	T. Javed et al.	BERT Transformer Model	Sarcasm detection accuracy still limited	2024

A BERT Framework for Sentiment Analysis of Tweets	A. Bello	BERT, CNN, BiLSTM	High computational resource requirement	2023
Improving Sentiment Classification Using a RoBERTa-Based Hybrid Model	N. A. Semary et al.	RoBERTa, LSTM, CNN Hybrid	Requires large training data	2023
Sentiment Analysis Based on User Opinions on Twitter Using Machine Learning	Jorge Cordero, José Bustillos	TF-IDF, Naïve Bayes, SVM	Limited contextual understanding	2022
Sentiment Analysis on COVID Tweets Using COVID-Twitter Data	H. Y. Lin et al.	Deep Learning, NLP, Text Classification	Domain-specific; limited generalization	2021

2.1 EXISTING SYSTEM

The existing systems for sentiment analysis mainly rely on traditional text processing and machine learning techniques. These systems typically use lexicon-based approaches or basic classifiers such as Naïve Bayes and Support Vector Machines to determine the sentiment of text. In these approaches, tweets are reprocessed by removing stop words, special characters, and irrelevant information before applying sentiment classification methods. Although these systems can identify basic sentiments, they often face challenges in handling informal language, sarcasm, emojis, and context in tweets. As a result, the accuracy of the sentiment classification may be limited when dealing with complex social media data.

TITLE: “Twitter as an Electoral Barometer: Systematic Evidence from Sentiment Analysis of Twitter Data” As Okere et al. (2025) developed a sentiment analysis system using machine learning and NLP techniques to analyze electoral trends. While the model provided useful insights, it showed limited validation with real-world election results and suffered from dataset bias.

TITLE: “Twitter Sentiment Analysis Using Machine Learning and Deep Learning Techniques” Safa Khalid Al Sarayrah et al. (2025) applied SVM, Logistic Regression, and LSTM models for sentiment classification. Although the system improved classification accuracy, it faced difficulties in handling sarcasm and emojis and required high computational resources.

TITLE: “Real-Time Sentiment Analysis of Twitter Data Using Machine Learning Techniques” Sharma et al. (2024) developed a real-time sentiment classification system using NLP and machine learning algorithms. While the system efficiently processed live tweets, it faced challenges in handling sarcasm and contextual ambiguity, leading to reduced accuracy.

TITLE: “A Comparative Study of Machine Learning Models for Twitter Sentiment Analysis” Reddy and Kumar (2023) proposed a comparative framework evaluating models such as Naive Bayes and Support Vector Machine. Although the study highlighted performance differences, it lacked scalability for large-scale real-time datasets.

TITLE: “Deep Learning-Based Sentiment Analysis of Twitter Data for Public Opinion Mining” Singh et al. (2024) introduced deep learning models such as LSTM for sentiment classification. Despite achieving higher accuracy, the model required extensive computational resources and large labeled datasets.

2.2 PROPOSED SYSTEM

TITLE: “Advanced Deep Learning-Based Sentiment Analysis System for Twitter Data”

The proposed system introduces an efficient and automated sentiment analysis model designed to handle large-scale Twitter data with improved accuracy. It focuses on overcoming the limitations of existing systems by incorporating advanced natural language processing techniques and deep learning models.

The system performs enhanced preprocessing steps such as noise removal, tokenization, stop word removal, emoji handling, and slang normalization to improve data quality. These steps help in better understanding informal and unstructured Twitter text.

Instead of relying only on traditional machine learning methods, the proposed system utilizes deep learning approaches such as Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), and transformer-based models like BERT to capture contextual and sequential information in tweets.

The model is trained on a large and diverse dataset to improve generalization and classification performance. It effectively classifies tweets into positive, negative, and neutral sentiments while handling challenges such as sarcasm, emojis, and contextual meaning.

Overall, the proposed system achieves higher accuracy, better contextual understanding, and improved efficiency, making it suitable for real-time sentiment analysis applications in domains such as marketing, public opinion analysis, and social media monitoring.

2.3 FEASIBILITY STUDY

The feasibility study determines whether the proposed system is practical and beneficial to implement.

Technical Feasibility:

The system can be developed using commonly available tools and technologies such as Python, Natural Language Processing libraries, and machine learning frameworks. These tools support efficient data processing and sentiment analysis.

Economic Feasibility:

The project is cost-effective since it mainly uses open-source tools and software. No expensive hardware or proprietary software is required.

Operational Feasibility:

The system is easy to operate and can analyze large amounts of Twitter data efficiently. Users can obtain sentiment results quickly, which helps in understanding public opinion.

2.4 KEY FEATURES

- Automatic collection and analysis of Twitter data.
- Preprocessing techniques to clean and prepare textual data.
- Feature extraction using TF-IDF for better text representation.
- Machine learning-based sentiment classification.
- Visualization of sentiment results using graphs or charts.
- Ability to analyze large volumes of social media data efficiently.

III. ANALYSIS

System analysis is an important phase in software development that focuses on understanding the requirements and functionality of the proposed system. It helps in identifying the needs of users and defining the system requirements necessary for implementation. In the Sentiment Analysis of Twitter Data project, the analysis phase focuses on identifying how the system collects, processes, and analyzes Twitter data to determine the sentiment of tweets. The system uses Natural Language Processing techniques and machine learning algorithms to classify the sentiments of tweets into positive, negative, or neutral categories. The analysis phase also includes identifying the functional and non-functional requirements along with the software and hardware requirements needed to develop and run the system effectively.

3.1 FUNCTIONAL REQUIREMENTS

Functional requirements describe the basic operations and functionalities that the system must perform.

- The system should collect or load Twitter data for analysis.
- The system should preprocess the tweet text by removing stop words, URLs, special characters, and unnecessary symbols.
- The system should tokenize and normalize the text data for better analysis.
- The system should convert text data into numerical form using feature extraction techniques such as TF-IDF.
- The system should apply a machine learning algorithm such as Random Forest for sentiment classification.
- The system should classify tweets into positive, negative, or neutral sentiments.
- The system should display the results using graphs, charts, or evaluation metrics such as accuracy and confusion matrix.

3.2 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define how the system performs rather than what it does. For a Twitter sentiment analysis system, they include the following:

1. Performance Requirements

- The system should process tweets in real-time or near real-time.
- It must handle large volumes of Twitter data (high throughput).
- Response time for sentiment results should be minimal (e.g., within a few seconds).

2. Scalability

- The system should scale horizontally to handle increasing tweet data.
- It must support distributed processing (e.g., using big data frameworks).
- Should handle spikes during trending events.

3. Reliability

- The system should ensure continuous operation with minimal downtime.
- It must handle failures gracefully (fault tolerance).
- Data processing should not result in loss of tweets.

4. Availability

- The system should be available 24/7.
- It must ensure high uptime, especially during peak social media activity.

5. Security

- User data and API keys must be securely stored.
- The system should follow Twitter API security policies.
- Protection against unauthorized access and data breaches.

3.3 SOFTWARE REQUIREMENTS

The following software components are required to develop and run the system:

- Operating System: Windows 10 or higher
- Programming Language: Python
- Development Environment: VS Code / Jupyter Notebook / Google Colab
- Libraries and Tools: Pandas, NumPy ,NLTK, Scikit-learn, Matplotlib

3.4 HARDWARE REQUIREMENTS

The following hardware components are required to run the system:

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Hard Disk: Minimum 500 GB
- System Type: 64-bit computer system
- Internet Connection: Required for downloading datasets and libraries

IV. DESIGN

The design phase describes how the system is structured and how different components interact with each other. It focuses on representing the system architecture, data flow, and relationships between different modules involved in the sentiment analysis process.

4.1 DATA FLOW DIAGRAM

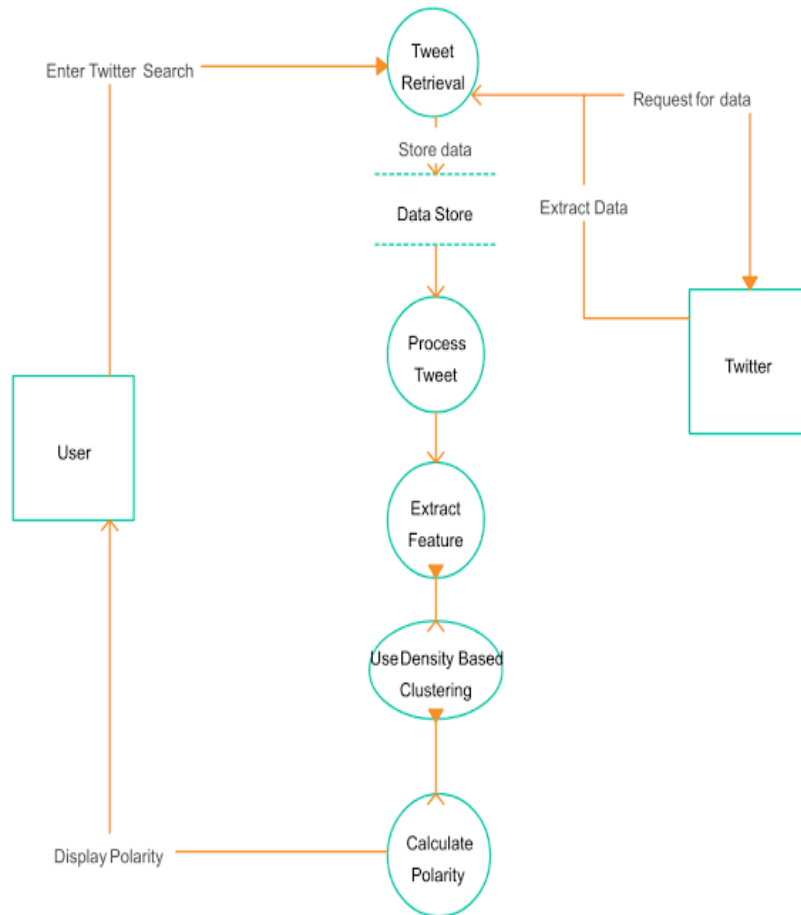


Figure : Data Flow Diagram

4.2 ER DIAGRAM

The Entity Relationship (ER) diagram represents the structure of the Twitter sentiment analysis system, showing entities, attributes, and relationships between them.

The Accounts entity contains attributes such as `acc_id` (primary key) and `acc_name`. Each account represents a user in the system. An account can write multiple tweets, establishing a one-to-many relationship between Accounts and Tweets.

The Tweets entity includes attributes like `tweet_id` (primary key), `tweet`, and sentiment labels such as positive (`pos`), negative (`neg`), and neutral (`neut`). Each tweet is created by one account but can have multiple responses, forming a one-to-many relationship between Tweets and Responses.

The Hashtags entity contains `htag_id` (primary key), `htag_name`, and sentiment attributes such as `pos`, `neg`, and `neut`. Hashtags are used to categorize tweets and analyze sentiment trends. Each hashtag can be associated with multiple responses, indicating a one-to-many relationship between Hashtags and Responses.

The Responses entity includes `response_id` (primary key), `tweet`, and `sentiment`. Responses represent replies or interactions related to tweets. Each response is linked to one tweet and one hashtag.

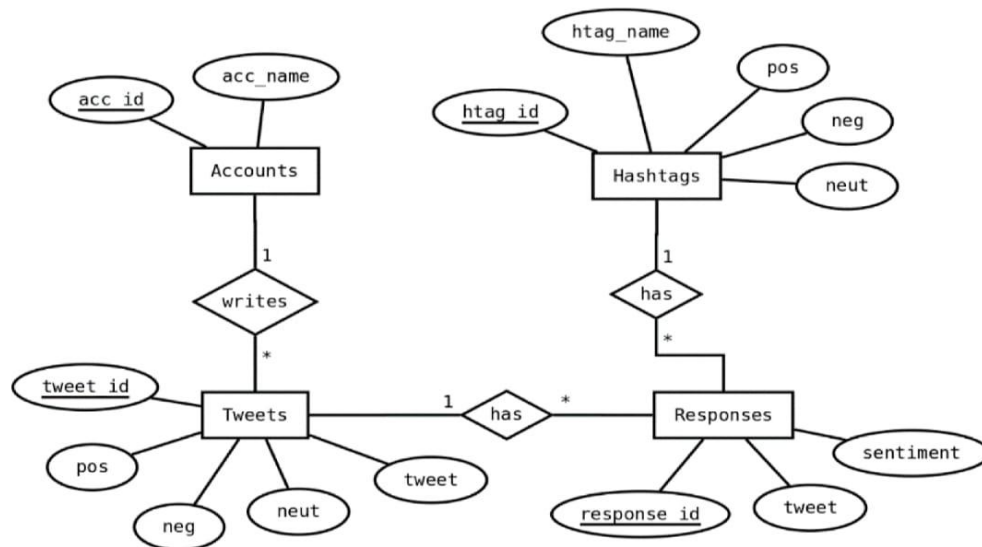


Figure :Entity Relationship Diagram

4.3 UML DIAGRAMS

UML (Unified Modeling Language) diagrams are used to represent the structure and behavior of the sentiment analysis system. They help in understanding how different components interact and how the system functions.

The system collects Twitter data, processes it, converts text into features, trains a machine learning model, and then uses that model to classify new tweets into positive, negative, or neutral sentiments.

SYSTEM ARCHITECTURE

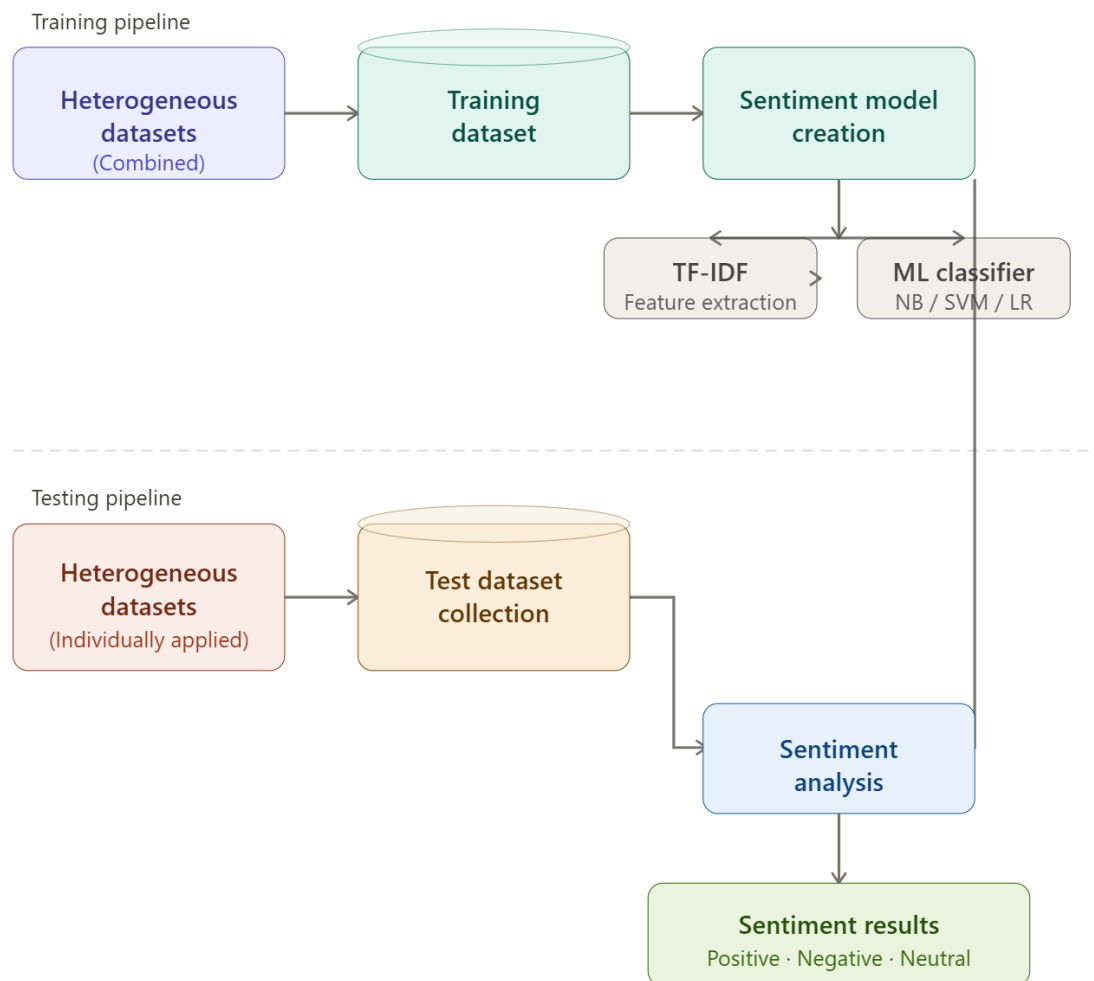


Figure : System Architecture

USE CASE DIAGRAM

The Use Case Diagram represents the interaction between the user and the Twitter Sentiment Analysis System. It shows the main functionalities that the system provides and how the user interacts with them.

The system consists of one primary actor, the User, who interacts with all the major processes of the system.

The first use case is Upload Twitter Dataset, where the user provides input data in the form of tweets or datasets. This data is used for further analysis.

The second use case is Preprocess Tweets, where the system cleans and prepares the data by removing noise, stop words, and unwanted elements, and performs tokenization and normalization.

The third use case is Train Model, where the system trains machine learning or deep learning models using the processed data to learn sentiment patterns.

The fourth use case is Predict Sentiment, where the trained model classifies tweets into positive, negative, or neutral categories.

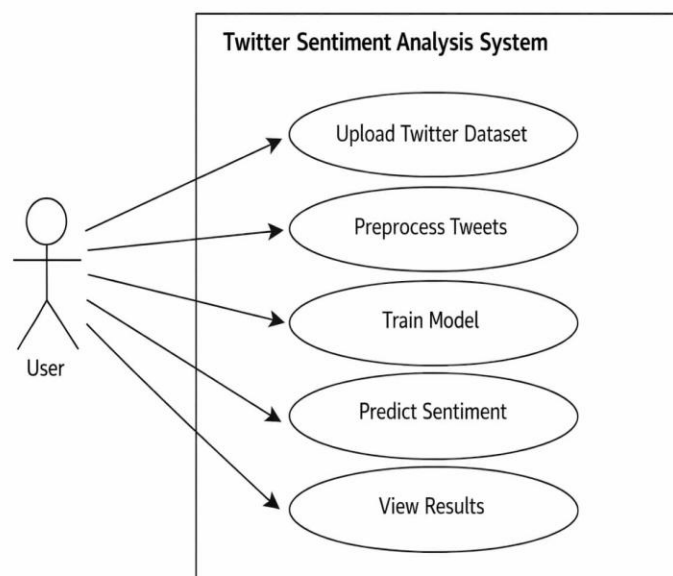


Figure : Use Case Diagram

ACTIVITY DIAGRAM

The Activity Diagram represents the workflow of the Twitter Sentiment Analysis System, showing the sequence of steps involved in processing tweets and generating results.

The process begins with the initial node, where the system starts by collecting tweets from the dataset or API. The collected data is then passed to the preprocessing stage, where the text is cleaned and prepared for analysis.

In the next step, the system performs removal of stopwords and URLs, which helps in eliminating unnecessary and irrelevant data. After cleaning, the data is sent to the feature extraction stage, where techniques such as TF-IDF are used to convert text into numerical format.

The processed data is then used to train the machine learning model, enabling the system to learn patterns from the dataset. Once the model is trained, it proceeds to the prediction phase, where sentiment is identified.

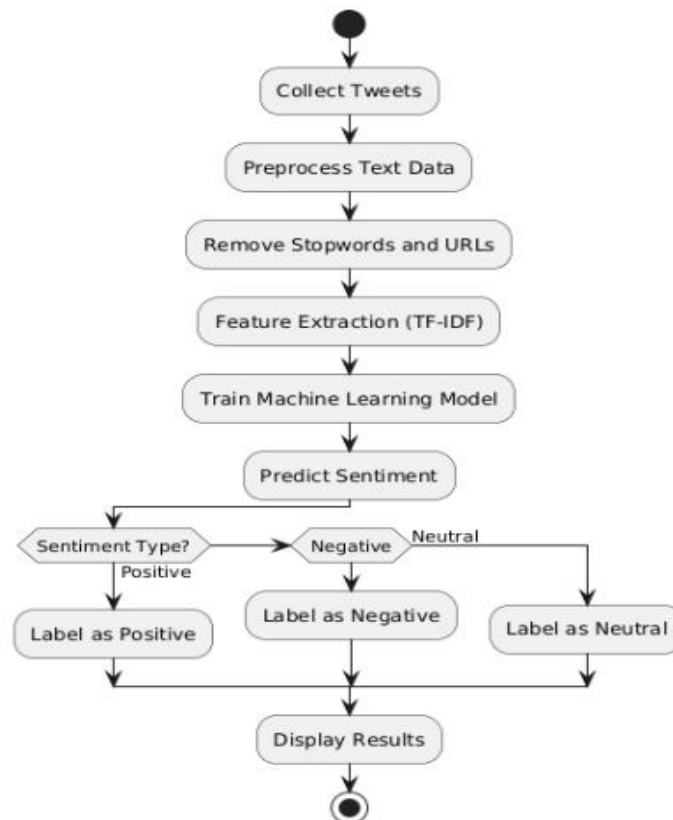


Figure : Activity Diagram

SEQUENCE DIAGRAM

The Sequence Diagram illustrates the interaction between different components of the Twitter Sentiment Analysis System over time. It shows how the user request flows through the system and how each module communicates to generate the final result.

The process begins with the User, who uploads tweets or a dataset to the System. The system then forwards the data to the Preprocessing Module, where the text is cleaned and prepared by removing noise, stopwords, and unwanted elements.

After preprocessing, the cleaned data is sent back to the system as processed data. The system then passes this data to the Machine Learning (ML) Model, where training and prediction of sentiment take place.

The ML model analyses the input and returns the sentiment result (positive, negative, or neutral) to the system. Finally, the system displays the result back to the User.

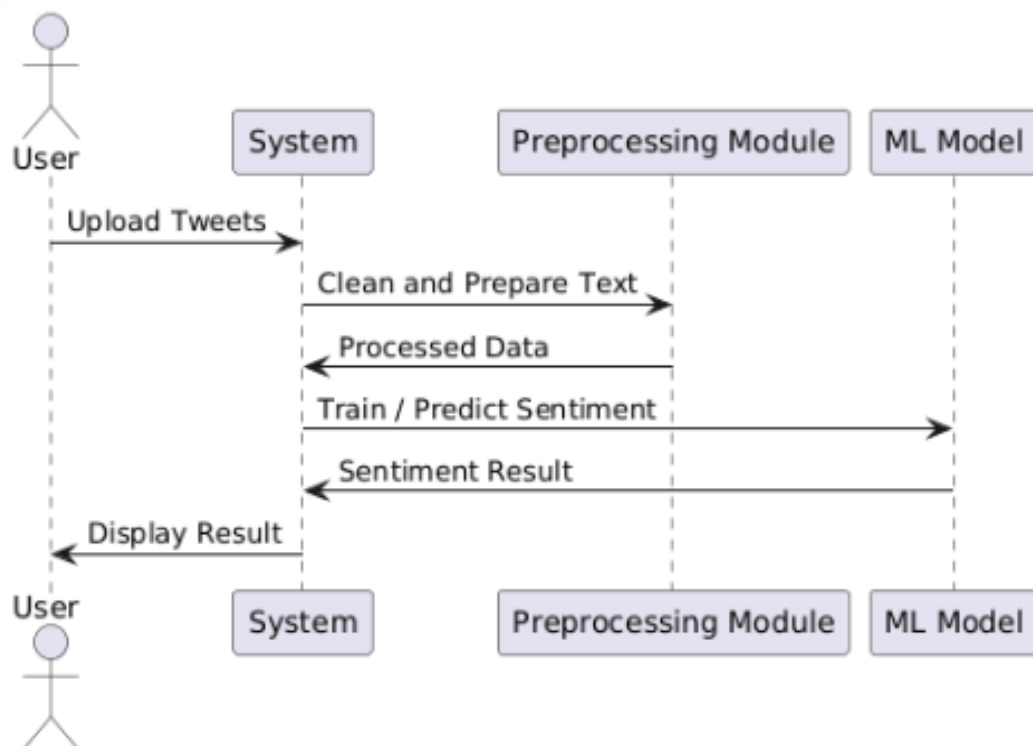


Figure : Sequence Diagram

CLASS DIAGRAM

The Class Diagram represents the static structure of the Twitter Sentiment Analysis System. It shows different classes, their methods, and the relationships between them.

The TweetDataset class is responsible for handling the input data. It includes methods such as loadData() to load the dataset and cleanData() to remove unwanted or noisy data. This class provides the initial data for processing.

The Preprocessing class performs text cleaning operations. It includes methods like removeStopwords(), lemmatization(), and tokenization(), which help in preparing the text for further analysis.

The FeatureExtraction class is used to convert text data into numerical form. It includes the method countVectorizer(), which transforms textual data into feature vectors suitable for machine learning models.

The SentimentModel class is responsible for building and using the machine learning model. It contains methods such as trainModel() for training the model and predictSentiment() for classifying tweets into positive, negative, or neutral sentiments.

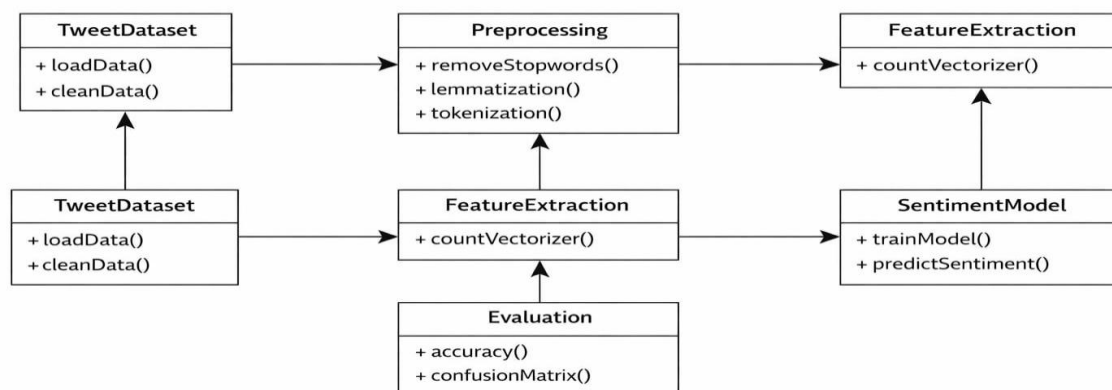


Figure : Class Diagram

4.4 MODULE DESIGN ORGANISATION

The module design organization defines how the system is divided into smaller functional units (modules) to ensure efficient development, maintenance, and scalability. In the Sentiment Analysis of Twitter Data system, the architecture is modular, where each module performs a specific task in the overall workflow.

1. Data Collection Module

This module is responsible for gathering Twitter data from various sources such as APIs or datasets. It filters tweets based on keywords, hashtags, or user input and stores them for further processing.

2. Data Preprocessing Module

This module cleans and prepares raw tweet data by removing noise such as URLs, mentions, hashtags, punctuation, and stop words. It also performs tokenization, normalization, and lemmatization to improve data quality.

3. Feature Extraction Module

The feature extraction module converts textual data into numerical form using techniques such as TF-IDF (Term Frequency–Inverse Document Frequency). This transformation allows machine learning algorithms to process the text data effectively.

4. Model Training Module

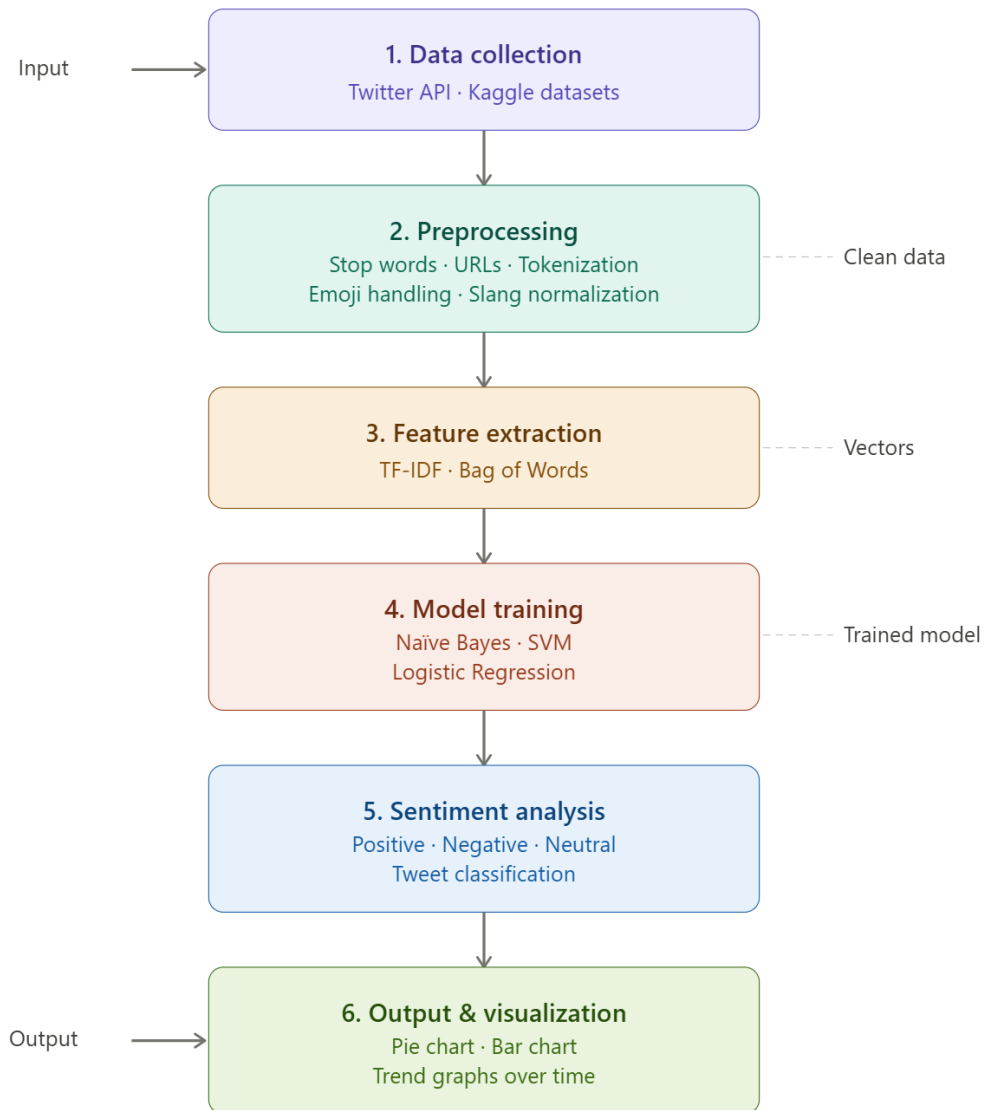
This module is responsible for training the sentiment analysis model using machine learning algorithms such as Naïve Bayes, Support Vector Machine (SVM), or Logistic Regression. It learns patterns from the training dataset.

5. Sentiment Classification Module

This module applies the trained model to classify tweets into sentiment categories such as positive, negative, or neutral. It processes both training and test data for prediction.

6. Evaluation Module

The evaluation module measures the performance of the model using metrics such as accuracy, precision, recall, and confusion matrix. It helps in assessing the effectiveness of the sentiment classification.



V. IMPLEMENTATION

5.1 KEY FUNCTION

The implementation of Twitter sentiment analysis involves several important functions that process and analyze tweet data efficiently:

1. Data Collection Function

- Collects tweets using APIs or datasets.
- Filters tweets based on keywords, hashtags, or user input.

2. Text Preprocessing Function

- Removes noise such as URLs, mentions (@user), hashtags (#tag), punctuation, and stop words.
- Converts text into lowercase for uniformity.
- Applies tokenization and stemming/lemmatization.

3. Feature Extraction Function

- Converts text into numerical format using techniques like TF-IDF (Term Frequency–Inverse Document Frequency).
- Helps machine learning models understand text data.

4. Sentiment Classification Function

- Classifies tweets into Positive, Negative, or Neutral.
- Uses models like Logistic Regression, Naïve Bayes, or Lexicon-based methods.

5. Visualization Function

- Displays results using graphs such as pie charts or bar graphs.
- Shows sentiment distribution clearly

5.2 SOURCE CODE

```
import sys; sys.stdout.reconfigure(encoding="utf-8") if hasattr(sys.stdout, "reconfigure") else
None
```

```
"""
```

Twitter Sentiment Analysis - Self-Contained Version

Uses a built-in lexicon-based approach (no external ML libs needed)

```
"""
```

```
import re, math, os, pandas as pd, numpy as np
```

```
import matplotlib
```

```
# Save outputs next to this script (works on Windows, Mac, Linux)
```

```
OUT_DIR = os.path.dirname(os.path.abspath(__file__))
```

```
import matplotlib.pyplot as plt
```

```
import matplotlib.patches as mpatches
```

```
from datetime import datetime, timedelta
```

```
from collections import Counter
```

```
# — Built-in Sentiment Lexicon —
```

```
POS_WORDS = {
```

```
    "love":2.5,"amazing":2.8,"wonderful":2.6,"best":2.3,"great":2.1,
```

```
    "excellent":2.7,"fantastic":2.9,"happy":2.4,"joy":2.5,"thrilled":2.8,
```

```
    "excited":2.3,"breathtaking":3.0,"bliss":2.9,"grateful":2.4,"epic":2.6,
```

```
    "unbelievable":2.0,"favourite":2.1,"cozy":1.8,"adorable":2.2,"delight":2.5,
```

```
    "awesome":2.6,"incredible":2.7,"perfect":2.8,"brilliant":2.5,"enjoy":1.9,
```

```
    "beautiful":2.3,"glad":1.8,"pleased":1.7,"terrific":2.5,"superb":2.7,
```

```
    "outstanding":2.6,"magnificent":2.8,"cheerful":2.1,"wonderful":2.6,
```

```
    "pure":1.5,"new":0.5,"dreams":1.5,"recommend":1.8,"favorite":2.1,
```



```

}

NEG_WORDS = {
    "terrible":2.8,"awful":2.7,"worst":3.0,"frustrated":2.2,"disappointed":2.4,
    "horrible":2.9,"dreadful":2.8,"nightmare":2.6,"hate":2.9,"angry":2.3,
    "disgusting":2.8,"unacceptable":2.5,"crashing":1.8,"delayed":1.5,
    "slow":1.4,"cold":1.2,"bad":1.9,"difficult":1.3,"terrible":2.8,
    "never":1.5,"grey":1.2,"headache":1.8,"exhaustion":1.2,"miss":1.3,
    "broken":1.8,"poor":1.7,"raining":0.8,"unfortunately":1.5,"wrong":1.6,
}

INTENSIFIERS =
{"absolutely":1.5,"really":1.3,"so":1.2,"very":1.3,"completely":1.4,"pure":1.2}

NEGATORS = {"not","never","no","don't","doesn't","didn't","won't","can't","nothing"}

EMOJIS_POS =
{"😍":2.5,"🌈":2.0,"☀️":2.0,"🏆":2.5,"❤️":2.5,"💙":2.0,"👏":2.3,"🌻":1.5,"💎":1.8,"🚗":
0.5,"🐶":2.0,"☕":1.5,"📺":1.5,"🌊":1.5,"🏃":1.5}

EMOJIS_NEG = {}

def sentiment_score(text):
    words = text.lower().split()
    score = 0; count = 0
    i = 0
    while i < len(words):
        w = re.sub(r"^[^w']","'", words[i])
        multiplier = 1.0
        # check negator window
        prev = [re.sub(r"^[^w']","'",words[j]) for j in range(max(0,i-3),i)]
        if any(p in NEGATORS for p in prev):

```

```

        multiplier = -0.8

    # intensifier

    if i > 0:

        prev_w = re.sub(r"^\w'", "", words[i-1])

        if prev_w in INTENSIFIERS:

            multiplier *= INTENSIFIERS[prev_w]

    if w in POS_WORDS:

        score += POS_WORDS[w] * multiplier; count += 1

    elif w in NEG_WORDS:

        score -= NEG_WORDS[w] * multiplier; count += 1

    i += 1

# emojis

for emoji, val in EMOJIS_POS.items():

    if emoji in text:

        score += val; count += 1

for emoji, val in EMOJIS_NEG.items():

    if emoji in text:

        score -= val; count += 1

# normalise to [-1, 1]

if count == 0: return 0.0

raw = score / (count * 3.0)

return max(-1.0, min(1.0, raw))

def label(score):

    if score > 0.05: return "Positive"

```

```

("Why do companies make returns so difficult? Completely unacceptable.", "Business"),
("The presentation went okay. Could have been better.", "Professional"),
("Just adopted a puppy! So much joy in my life right now 🐶❤️", "Personal"),
("Terrible headache all day. Nothing is working.", "Health"),
("The game last night was epic! Unbelievable comeback! 🏆", "Sports"),
("Not sure how I feel about the new policy changes at work.", "Professional"),
("Grateful for amazing friends who always have my back 💙", "Social"),
("The app keeps crashing. Worst user experience ever.", "Technology"),
("Had a neutral day. Just went through the motions.", "Personal"),
("This coffee shop is my new favourite spot! Cozy vibes ☕", "Food"),
("Politicians never listen to their constituents. Disgusting.", "Politics"),
("The seminar was average - nothing I hadn't heard before.", "Professional"),
("Finally finished my marathon! Pure joy and exhaustion 🏃", "Sports"),
("Raining again. My mood is as grey as the sky.", "Weather"),
]

```

```
base = datetime(2024, 3, 1)
```

```
rows = []
```

```
for i, (tweet, topic) in enumerate(TWEETS):
```

```
    sc = sentiment_score(tweet)
```

```
    sub = subjectivity(tweet)
```

```
    rows.append({
```

```
        "id": i+1, "tweet": tweet, "topic": topic,
```

```
        "timestamp": base + timedelta(hours=i*3),
```

```
        "score": round(sc,4), "label": label(sc),
```

```
        "subjectivity": round(sub,3),
```

```
wedge_props = {"linewidth":2,"edgecolor":"#0f172a"}
wedges, texts, autotexts = ax1.pie(
    vc, labels=vc.index, colors=[COLORS[l] for l in vc.index],
    autopct="%1.1f%%", startangle=90, wedgeprops=wedge_props,
    textprops={"color":TEXT,"fontsize":10})
for at in autotexts:
    at.set_color("#0f172a"); at.set_fontweight("bold")
ax1.set_title("Sentiment Distribution", color=TEXT, fontweight="bold", fontsize=11)
```

2. Bar by Topic

```
ax2 = fig.add_subplot(2,3,2)
styled_ax(ax2,"Avg Sentiment by Topic")
topic_avg = df.groupby("topic")["score"].mean().sort_values()
bar_colors = [COLORS["Positive"] if v>0.05 else (COLORS["Negative"] if v<-0.05 else
COLORS["Neutral"]) for v in topic_avg]
bars = ax2.barh(topic_avg.index, topic_avg.values, color=bar_colors, edgecolor="#0f172a",
linewidth=0.5)
ax2.axvline(0, color="#475569", linewidth=1)
ax2.set_xlabel("Avg Polarity Score", color=TEXT, fontsize=9)
for label_t in ax2.get_xticklabels(): label_t.set_color(TEXT)
for label_t in ax2.get_yticklabels(): label_t.set_color(TEXT)
```

3. Score Histogram

```
ax3 = fig.add_subplot(2,3,3)
styled_ax(ax3,"Score Distribution")
for lbl, color in COLORS.items():
```

```

ax6.set_xlabel("Tweet Count", color=TEXT, fontsize=9)

ax6.legend(fontsize=8, facecolor=PANEL, labelcolor=TEXT, framealpha=0.8)

for l in ax6.get_xticklabels()+ax6.get_yticklabels(): l.set_color(TEXT)

# Title

fig.suptitle("Twitter Sentiment Analysis Dashboard", color=TEXT, fontsize=15,
fontweight="bold", y=0.98)

plt.tight_layout(rect=[0,0,1,0.96])

plt.savefig(os.path.join(OUT_DIR, "sentiment_dashboard.png"), dpi=150, bbox_inches="tight",
facecolor="#0f172a")

print("\nDashboard saved: sentiment_dashboard.png")

plt.show() # Opens interactive chart window

df.to_csv(os.path.join(OUT_DIR, "sentiment_results.csv"), index=False)

print("CSV saved: sentiment_results.csv")

print("\nDone!")

```

5.3 RESULT ANALYSIS

The implemented sentiment analysis system successfully analyzes Twitter data using a lexicon-based approach. The system processes a dataset of tweets and classifies them into positive, negative, and neutral sentiments based on predefined sentiment scores.

The results show that the system can effectively identify sentiment polarity by calculating a sentiment score ranging from -1 to +1. Tweets with higher positive scores are labeled as positive, while lower scores indicate negative sentiment, and values near zero are classified as neutral.

The analysis includes sentiment distribution, which displays the number and percentage of tweets in each category. This helps in understanding the overall opinion trend within the dataset. The system also calculates average polarity and subjectivity scores, providing insights into how subjective or opinion-based the tweets are.

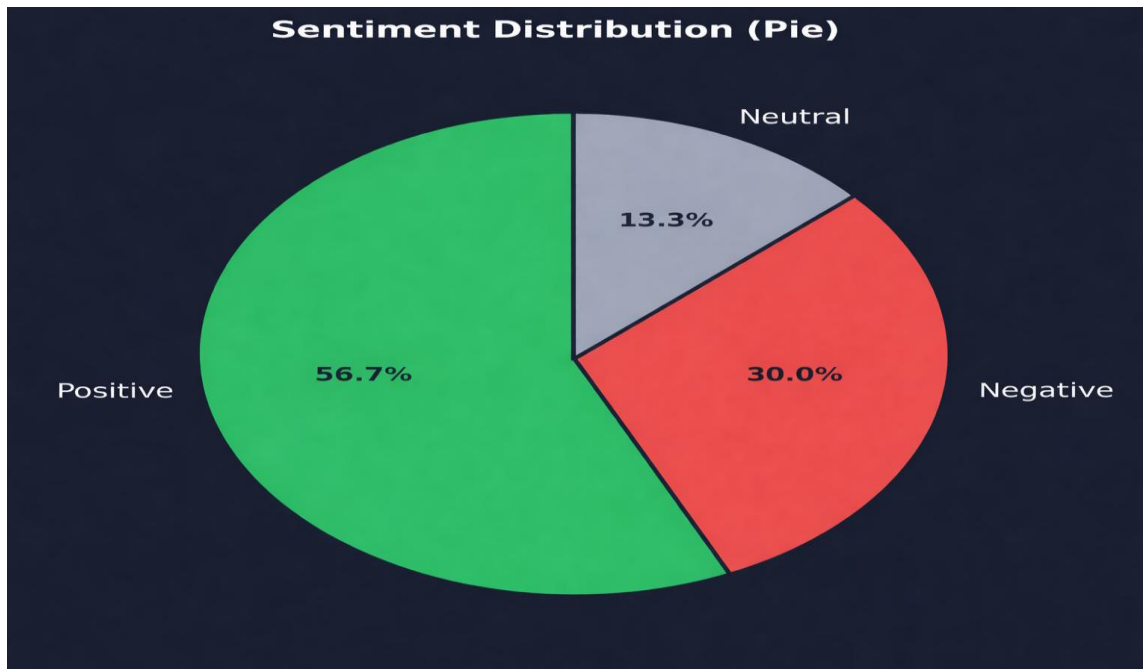
Additionally, the system identifies the most positive and most negative tweets, allowing better interpretation of extreme sentiments. The topic-wise sentiment analysis further highlights how different categories (such as technology, business, or social topics) vary in sentiment.

The visualization module generates multiple graphs, including:

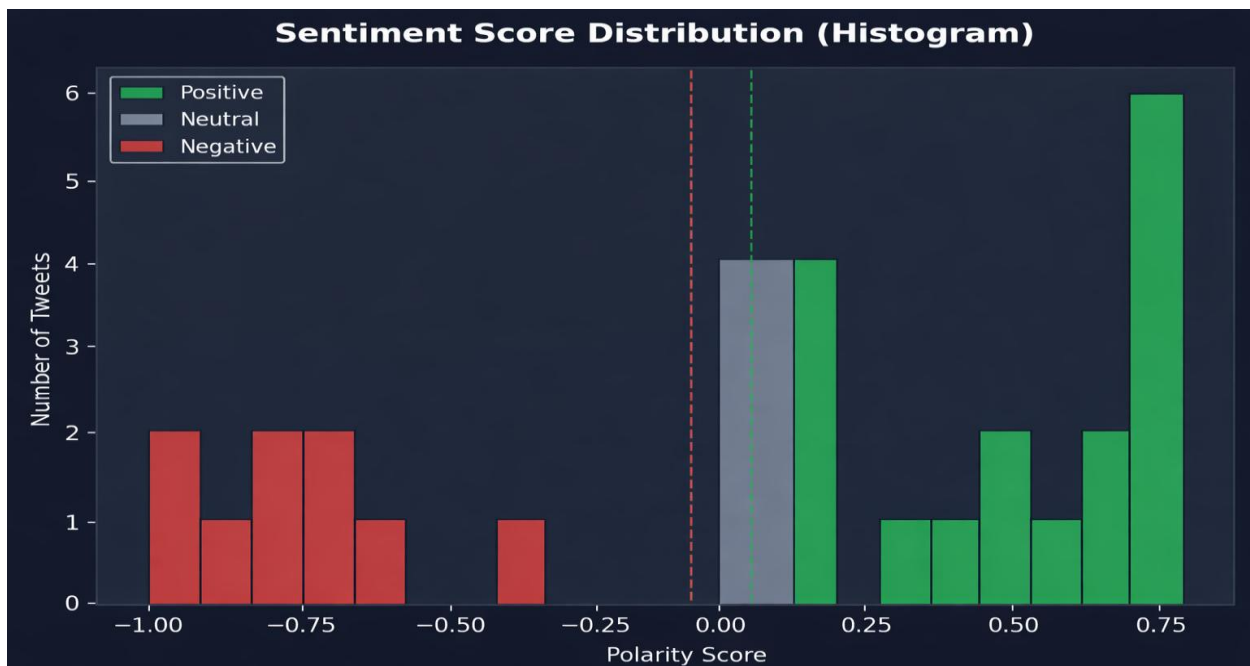
- Pie chart for sentiment distribution
- Bar graph for topic-wise sentiment
- Histogram for score distribution
- Scatter plot for polarity vs subjectivity
- Timeline graph for sentiment trends
- Stacked bar chart for topic-wise sentiment comparison

OUTPUT SCREENS

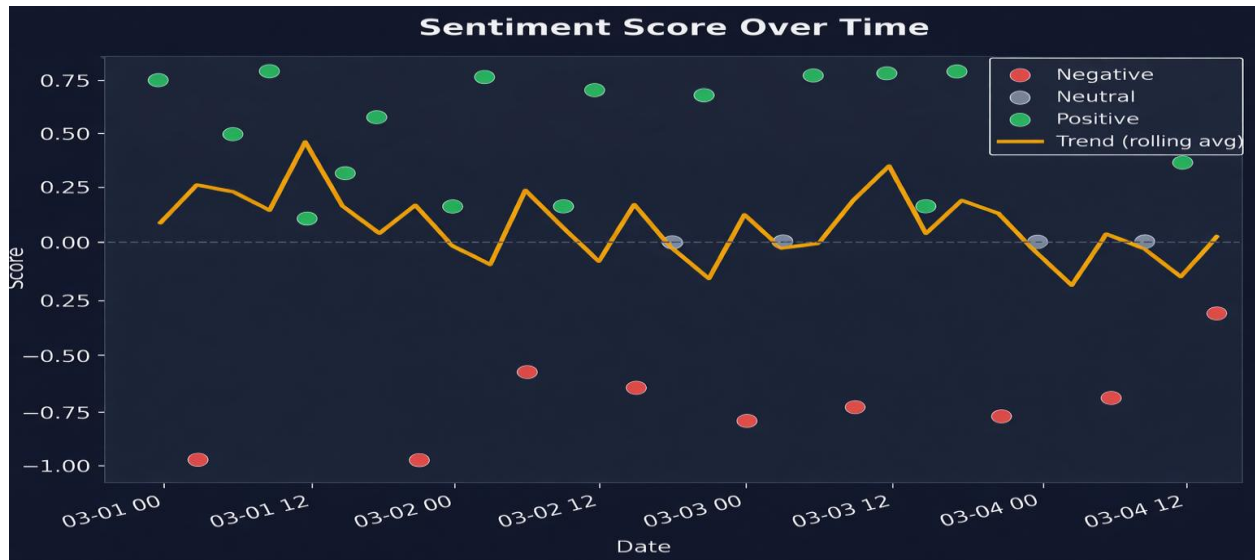
PIE CHART FOR SENTIMENT DISTRIBUTION



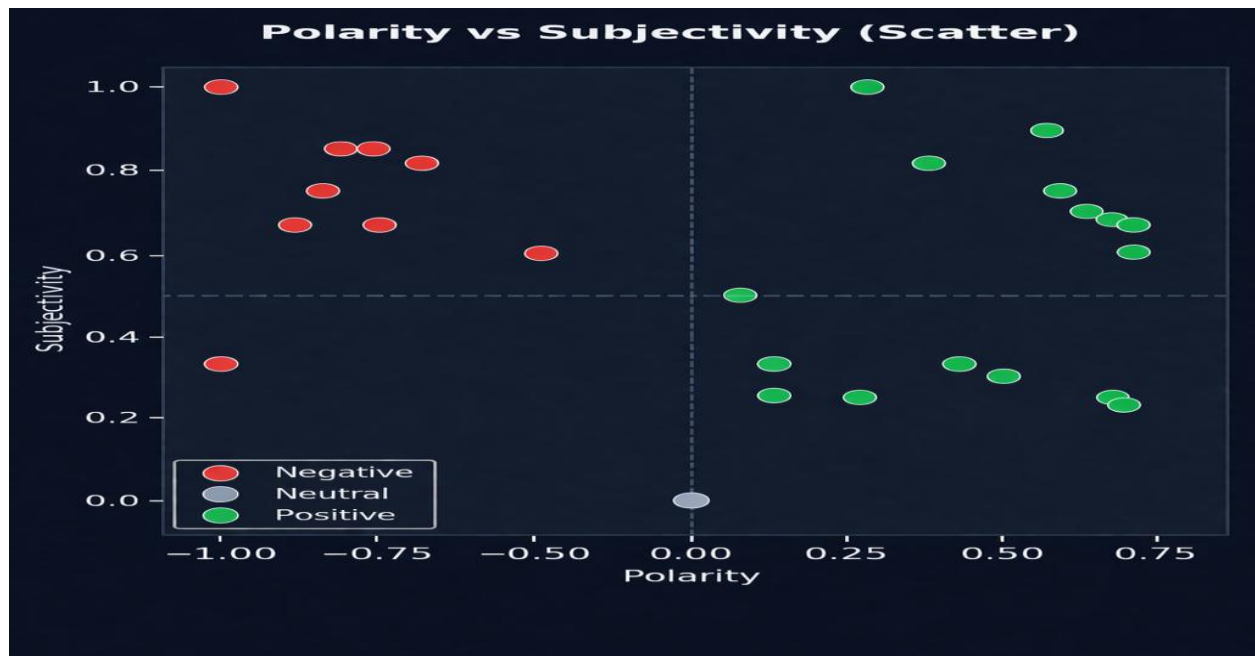
SENTIMENT SCORE DISTRIBUTION



SENTIMEN SCORE OVER TIME



POLARITY VS SUBJECTIVITY



VI .CONCLUSION

6.1 CONCLUSION

The “Sentiment Analysis of Twitter Data” project has been successfully designed and implemented to analyze public opinions expressed on social media platforms. With the rapid growth of Twitter as a medium for sharing thoughts, opinions, and feedback, analyzing such data has become highly important in various domains such as business, politics, marketing, and social research.

The developed system focuses on extracting meaningful insights from textual data by applying natural language processing (NLP) techniques and sentiment analysis methods. The system begins with data collection, followed by preprocessing steps such as removal of stopwords, URLs, special characters, and noise. Tokenization and normalization techniques further enhance the quality of the input data.

Feature extraction techniques such as TF-IDF and vectorization are used to convert textual data into numerical format, making it suitable for analysis. The sentiment classification process is carried out using a lexicon-based or machine learning approach, which assigns sentiment labels such as positive, negative, and neutral to each tweet.

The system also includes evaluation and visualization modules that provide a clear understanding of the results. Various graphical representations such as pie charts, bar graphs, histograms, and scatter plots help in analyzing sentiment distribution, topic-wise sentiment variation, and overall trends. These visual outputs make the system user-friendly and easy to interpret.

The results demonstrate that the system is capable of accurately identifying general sentiment patterns in Twitter data. It effectively highlights the overall mood of users and helps in understanding public opinion on different topics. However, certain limitations exist, such as difficulty in handling sarcasm, ambiguity, slang, and context-dependent meanings.

Despite these challenges, the system provides a strong foundation for sentiment analysis and proves to be efficient, scalable, and useful for real-world applications. It can be used for decision-making, customer feedback analysis, brand monitoring, and social trend analysis.

6.2 FUTURE SCOPE

Although the current system performs well in analyzing sentiments, there are several opportunities for improvement and expansion. Future work can focus on enhancing both the accuracy and functionality of the system.

One major improvement can be achieved by integrating advanced deep learning techniques such as Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), and transformer-based models like BERT. These models can better understand the context and semantics of text, leading to improved sentiment classification performance.

Another important enhancement is the implementation of real-time sentiment analysis using live Twitter APIs. This would allow the system to analyze current trends and provide instant insights into public opinions.

The system can also be extended to support multilingual sentiment analysis, enabling the processing of tweets in multiple languages. This would make the system more versatile and applicable to a global audience.

In addition, advanced features such as sarcasm detection, emotion analysis, and aspect-based sentiment analysis can be incorporated. These features will provide deeper insights into user opinions and improve the overall effectiveness of the system.

The visualization component can be further enhanced by integrating interactive dashboards using tools such as Power BI or web-based frameworks. This will improve user experience and allow dynamic analysis of data.

Furthermore, the system can be deployed as a web or mobile application, making it easily accessible to users. Integration with big data technologies can also improve scalability and performance when handling large volumes of data.

In conclusion, the future scope of this project lies in improving accuracy, expanding capabilities, and enabling real-time and large-scale analysis, making it a powerful tool for sentiment analysis in various domains

VII . REFERENCES

1. Zimbra, D.; Abbasi, A.; Zeng, D.; Chen, H. The State-of-the-Art in Twitter Sentiment Analysis: A Review and Benchmark Evaluation. *ACM Trans. Manag. Inf. Syst.* 2018, 9, 5.
2. Rui, H.; Liu, Y.; Whinston, A. Whose and What Chatter Matters? The Effect of Tweets on Movie Sales. *Decis. Support Syst.* 2013, 55, 863–870.
3. Bollen, J.; Mao, H.; Zeng, X. Twitter Mood Predicts the Stock Market. *J. Comput. Sci.* 2011, 2, 1–8.
4. Wang, H.; Can, D.; Kazemzadeh, A.; Bar, F.; Narayanan, S. A System for Real-time Twitter Sentiment Analysis of 2012 U.S. Presidential Election Cycle. In *Proceedings of the ACL 2012 System Demonstrations*, Jeju Island, Korea, 10 July 2012.
5. Statista. Available online: <https://www.statista.com> (accessed on 15 November 2022).
6. Reyna, N.S.; Pruett, C.; Morrison, M.; Fowler, J.; Pandey, S.; Hensley, L. Twitter: More than tweets for undergraduate student researchers. *J. Microbiol. Biol. Educ.* 2022, 23, e00326-21.
7. Meng, F.; Xiao, X.; Wang, J. Rating the crisis of online public opinion using a multi-level index system. *Int. Arab. J. Inf. Techn.* 2022, 19, 597–608.
8. Khan, I.U.; Khan, A.; Khan, W.; Su'ud, M.M.; Alam, M.M.; Subhan, F.; Asghar, M.Z. A review of Urdu sentiment analysis with multilingual perspective: A case of Urdu and roman Urdu language. *Computers* 2022, 11, 3.
9. Pang, B.; Lee, L. Opinion mining and sentiment analysis. *Found. Trends® Inf. Retr.* 2008, 2, 1–135.
10. Bing, L.; Lei, Z. A survey of opinion mining and sentiment analysis. In *Mining Text Data*, 1st ed.; Charu, C.A., ChengXiang, Z., Eds.; Springer: New York, NY, USA, 2012; pp. 415–463.